

DETERMINANTS



The background of the slide features a collection of realistic water droplets of various sizes. Some droplets are large and prominent, while others are small and scattered. They are rendered with soft shadows and highlights, giving them a three-dimensional appearance. The droplets are primarily located on the right side of the slide, with a few smaller ones extending towards the center.**Definition:**

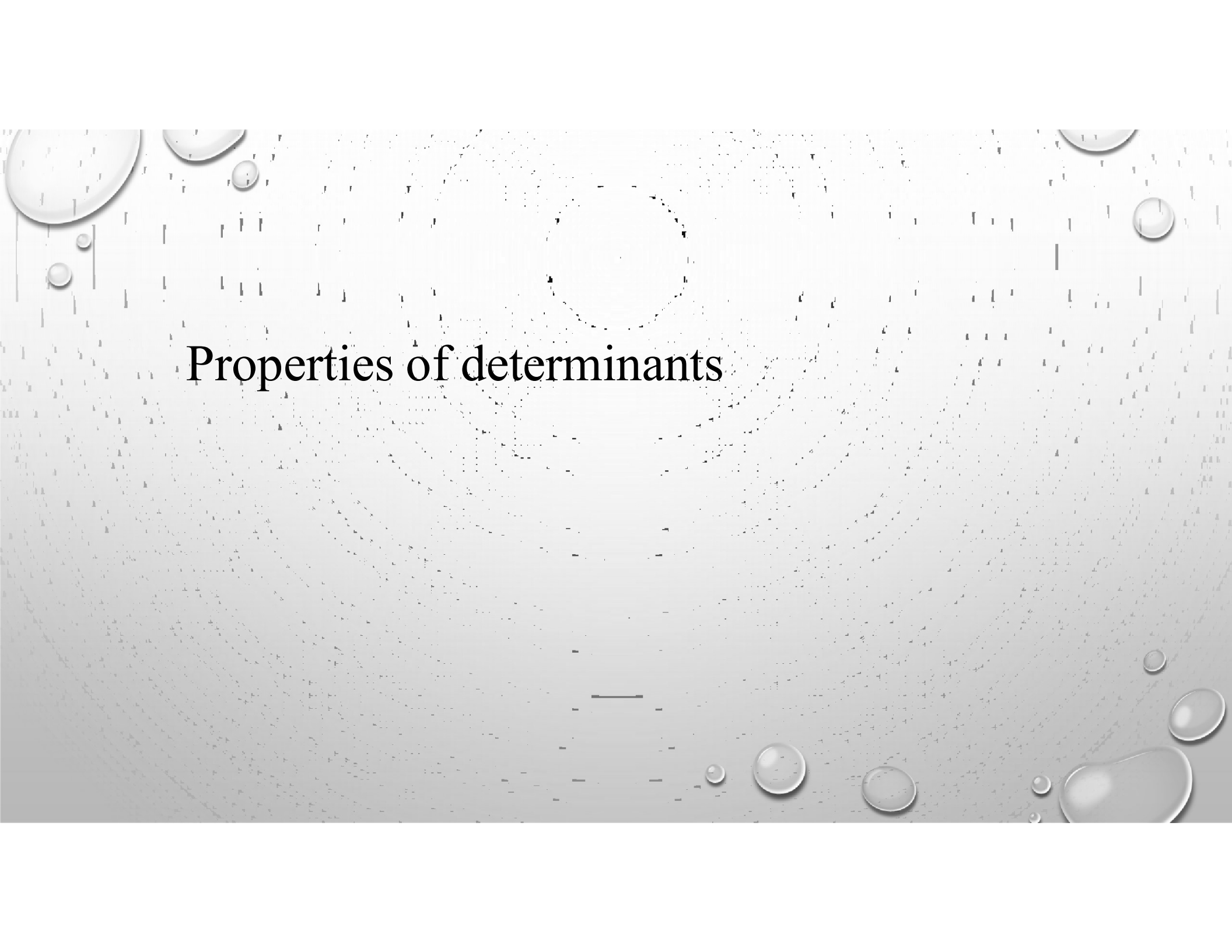
The determinant of a square matrix A of order n (with real or complex elements) is a real or complex number associated with matrix A of order n .

Determinant of a 2×2 Matrix

Determinant of a 2×2 matrix is defined as the difference of product of their main diagonal entries and second diagonal entries.

$$\begin{aligned} \text{i.e. } A &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\ |A| &= \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = (1 \times 4) - (3 \times 2) \\ &= 4 - 6 = -2 \end{aligned}$$



The background of the slide features a light gray gradient with a subtle, faint grid pattern. Scattered across the surface are numerous water droplets of varying sizes, some appearing as simple circles and others with more complex, realistic shading and highlights, giving the impression of a wet surface.

Properties of determinants

1st Property:

If we change the rows into columns and columns into rows, the value of the determinant will remain unaltered.

$$|A| = |A^t|$$

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Ex:

Let $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & 2 \end{bmatrix}$, verify that-

$$|A| = |A^t|.$$

Sol:

$$|A| = \begin{vmatrix} 1 & 0 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & 2 \end{vmatrix} \quad \text{expanding by } R_1$$

We get.

$$|A| = 1 \begin{vmatrix} 1 & 2 \\ 3 & 2 \end{vmatrix} - 0 \begin{vmatrix} 2 & 2 \\ 4 & 2 \end{vmatrix} + 3 \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix}$$

$$= 1(2 - 6) - 0(4 - 8) + 3(6 - 4)$$

$$= 1(-4) - 0 + 3(2)$$

$$= -4 + 6$$

$$|A| = 2$$

Now for RHS. $A^t = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 3 & 2 & 2 \end{bmatrix}$

$$|A^t| = \begin{vmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 3 & 2 & 2 \end{vmatrix} \text{ expanding by } R_1$$

we get.

$$\begin{aligned} |A^t| &= 1 \begin{vmatrix} 1 & 3 \\ 2 & 2 \end{vmatrix} - 2 \begin{vmatrix} 0 & 3 \\ 3 & 2 \end{vmatrix} + 4 \begin{vmatrix} 0 & 1 \\ 3 & 2 \end{vmatrix} \\ &= 1(2 - 6) - 2(0 - 9) + 4(0 - 3) \\ &= 1(-4) - 2(-9) + 4(-3) \\ &= -4 + 18 - 12 = -16 + 18 = 2 \end{aligned}$$

Hence, $|A| = |A^t|.$

2nd Property:

If we change any two rows or any two columns, then the sign of the determinant is changed but the value of the determinant will not be altered.

If we change second row by third row

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_2 & c_3 \end{vmatrix} = - \begin{vmatrix} a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

If we change first column by third column

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_2 & c_3 \end{vmatrix} = - \begin{vmatrix} c_1 & b_1 & a_1 \\ c_2 & b_2 & a_2 \\ c_3 & b_3 & a_3 \end{vmatrix}$$

Ex: Let $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & 2 \end{bmatrix}$

Sol: -

Interchange any two rows,
Say second and third, we get.

$$B = \begin{bmatrix} 1 & 0 & 3 \\ 4 & 3 & 2 \\ 2 & 1 & 2 \end{bmatrix}$$

Now prove that $|A| = -|B|$

So, first we find $LHS = |A|$.

$$|A| = \begin{vmatrix} 1 & 0 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & 2 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 2 & 2 \\ 3 & 2 \end{vmatrix} - 0 \begin{vmatrix} 2 & 2 \\ 4 & 2 \end{vmatrix} + 3 \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix}$$

$$= 1(2 - 6) - 0(4 - 8) + 3(6 - 4)$$

$$= 1(-4) - 0 + 3(2)$$

$$= -4 + 6$$

$$= 2$$

$$|B| = \begin{vmatrix} 1 & 0 & 3 \\ 4 & 3 & 2 \\ 2 & 1 & 2 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} - 0 \begin{vmatrix} 4 & 2 \\ 2 & 2 \end{vmatrix} + 3 \begin{vmatrix} 4 & 3 \\ 2 & 1 \end{vmatrix}$$

$$= 1(6-2) - 0 + 3(4-6)$$

$$= 4 - 0 - 6$$

$$= -2$$

Hence,

$$|A| = -|B|$$

SIMILARLY, interchange
any two columns says
second and third. we
get

$$C = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 2 & 1 \\ 4 & 2 & 3 \end{bmatrix} \text{ Now, } |C| = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 2 & 1 \\ 4 & 2 & 3 \end{vmatrix}$$

$$|C| = 1 \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} - 3 \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix} + 0 \begin{vmatrix} 2 & 2 \\ 4 & 2 \end{vmatrix}$$

$$= 1(6-2) - 3(6-4) + 0$$

$$= 4 - 3(2)$$

$$= 4 - 6 = -2$$

$$|C| = -|A|$$

3rd Property:

If Two Rows or two Columns are identical , the value of the determinant is zero.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \end{vmatrix} = 0 = \begin{vmatrix} a_1 & b_2 & b_2 \\ a_1 & b_2 & b_2 \\ a_1 & b_2 & b_2 \end{vmatrix}$$

Ex: I_f $A = \begin{bmatrix} 1 & 3 & 0 \\ 4 & 3 & 2 \\ 1 & 3 & 0 \end{bmatrix}$, Show that-

$$|A| = 0$$

Sol:- In matrix A two rows are identical, say 1st and 3rd.

$$|A| = \begin{vmatrix} 1 & 3 & 0 \\ 4 & 3 & 2 \\ 1 & 3 & 0 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 3 & 2 \\ 3 & 0 \end{vmatrix} - 3 \begin{vmatrix} 4 & 2 \\ 1 & 0 \end{vmatrix} + 0 \begin{vmatrix} 4 & 3 \\ 1 & 3 \end{vmatrix}$$

$$= 1(0-6) - 3(0-2) + 0(12-3)$$

$$= -6 + 6 + 0$$

$$|A| = 0 \quad \text{Hence proved.}$$

Example:-

Let, $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 0 & 2 \\ 4 & 2 & 4 \end{bmatrix}$, Show that $|A| = 0$

Sol:-

In matrix A, two columns are identical says 1st and 3rd.

$$|A| = \begin{vmatrix} 1 & 3 & 1 \\ 2 & 0 & 2 \\ 4 & 2 & 4 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 0 & 2 \\ 2 & 4 \end{vmatrix} - 3 \begin{vmatrix} 2 & 2 \\ 4 & 4 \end{vmatrix} + 1 \begin{vmatrix} 2 & 0 \\ 4 & 2 \end{vmatrix}$$

$$= 1(0 - 4) - 3(8 - 8) + 1(4 - 0)$$

$$= -4 - 0 + 4$$

$$|A| = 0 \quad \text{Hence proved.}$$

4th Property:

If the same constant multiplies each element in a row or a column, the value of the determinant is multiplied by that constant.

$$\begin{vmatrix} ma_1 & nb_1 & c_1 \\ ma_2 & nb_2 & c_2 \\ ma_3 & nb_3 & c_3 \end{vmatrix} = mn \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Ex:

Show that $\begin{vmatrix} 5 & 10 & 15 \\ 2 & 3 & 7 \\ 4 & 0 & 1 \end{vmatrix}$ is equal to

$$5 \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 7 \\ 4 & 0 & 1 \end{vmatrix}.$$

Sol:-

$$|A| = \begin{vmatrix} 5 & 10 & 15 \\ 2 & 3 & 7 \\ 4 & 0 & 1 \end{vmatrix} \text{ expanding by } R_1$$

$$= 5 \begin{vmatrix} 3 & 7 \\ 0 & 1 \end{vmatrix} - 10 \begin{vmatrix} 2 & 7 \\ 4 & 1 \end{vmatrix} + 15 \begin{vmatrix} 2 & 3 \\ 4 & 0 \end{vmatrix}$$

$$= 5(3-0) - 10(2-28) + 15(0-12)$$

$$= 15 - 10(-26) + 15(-12)$$

$$= 15 + 260 - 180 = 275 - 180$$

$$|A| = +275 - 180 = 95 = |A|$$

Now RHS:

$$5|A| = 5 \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 7 \\ 4 & 0 & 1 \end{vmatrix}$$

$$= 5 \left\{ 1 \begin{vmatrix} 3 & 7 \\ 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 7 \\ 4 & 1 \end{vmatrix} + 3 \begin{vmatrix} 2 & 3 \\ 4 & 0 \end{vmatrix} \right\}$$

$$= 5 \left\{ 1(3-0) - 2(2-28) + 3(0-12) \right\}$$

$$= 5 \{ 3 - 2(-26) + 3(-12) \}$$

$$= 5 \{ 3 + 52 - 36 \}$$

$$= 5(55 - 36)$$

$$= 5(19)$$

$$5|A| = 95$$

$$\text{Hence, } |A| = 5|A|$$

5th Property:

If each element of a row or a column of a determinant consists of two terms, the determinant is expressed as the sum of two determinants.

$$\begin{vmatrix} a_1 + p_1 & b_1 & c_1 \\ a_2 + p_2 & b_2 & c_2 \\ a_3 + p_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} p_1 & b_1 & c_1 \\ p_2 & b_2 & c_2 \\ p_3 & b_3 & c_3 \end{vmatrix}$$

Example: If

$$A = \begin{bmatrix} 18 & 5 & 1 \\ 22 & 7 & 2 \\ 19 & 9 & 3 \end{bmatrix}, B = \begin{bmatrix} 18 & 5 & 1 \\ 18 & 7 & 2 \\ 18 & 9 & 3 \end{bmatrix}, C = \begin{bmatrix} 0 & 5 & 1 \\ 4 & 7 & 2 \\ 1 & 9 & 3 \end{bmatrix}, \text{ then show}$$

that $|A| = |B| + |C|$

$$\begin{aligned} \text{Solution: L.H.S.} &= |A| = \begin{vmatrix} 18 & 5 & 1 \\ 22 & 7 & 2 \\ 19 & 9 & 3 \end{vmatrix} \\ &= \begin{vmatrix} 18+0 & 5 & 1 \\ 18+4 & 7 & 2 \\ 18+1 & 9 & 3 \end{vmatrix} \\ &= \begin{vmatrix} 18 & 5 & 1 \\ 18 & 7 & 2 \\ 18 & 9 & 3 \end{vmatrix} + \begin{vmatrix} 0 & 5 & 1 \\ 4 & 7 & 2 \\ 1 & 9 & 3 \end{vmatrix} \\ &= |B| + |C| = \text{R.H.S.} \end{aligned}$$

Statement :

If A and B are square matrices of the same order , then

$$|AB| = |A| \cdot |B|$$

Example: Let $A = \begin{bmatrix} l & m \\ -m & l \end{bmatrix}$ and $B = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then show that $|AB| = |A| \cdot |B|$

Work for the students



Statement : If A is a square matrix of order n , then $|kA| = k^n |A|$, where k is any scalar.

Exp: If $A = \begin{bmatrix} 5 & 4 \\ 7 & 8 \end{bmatrix}$ is a matrix of order 2. Show that $|5A| = 5^2 |A|$

Sol:

L.H.S

$$5A = 5 \begin{bmatrix} 5 & 4 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 35 & 40 \end{bmatrix}$$

$$|5A| = \begin{vmatrix} 25 & 20 \\ 35 & 40 \end{vmatrix}$$

$$|5A| = 25 \times 40 - 20 \times 35 = 1000 - 700 = 300$$

R.H.S

$$|A| = \begin{vmatrix} 5 & 4 \\ 7 & 8 \end{vmatrix} = 40 - 28 = 12$$

$$5^2 |A| = 25 \times 12 = 300$$

L.H.S = R.H.S proved

Statement: If A is a square matrix order n , then $|A^k| = |A|^k$.

Exp: If $A = \begin{bmatrix} 5 & 4 \\ 7 & 8 \end{bmatrix}$ is a square matrix, then show that $|A^3| = |A|^3$.

Proof:

L.H.S

$$A^3 = \begin{bmatrix} 5 & 4 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 25 + 28 & 20 + 32 \\ 35 + 56 & 28 + 64 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 7 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 53 & 52 \\ 91 & 92 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 265 + 364 & 212 + 416 \\ 455 + 644 & 364 + 736 \end{bmatrix}$$

$$= \begin{bmatrix} 629 & 628 \\ 1099 & 1100 \end{bmatrix}$$

$$|A^3| = 691,900 - 690,172 = 1728$$

R.H.S

$$|A| = 12$$

$$|A|^3 = 12^3 = 1728$$

L.H.S = R.H.S Proved



THANK YOU
FOR YOUR ATTENTION
ANY QUESTIONS?

